

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next Item →

1.

You provide access to your model's source code and document the data sources you used for training an AI model. This is an example of:

1 / 1 point

☐ Explainability

☐ Interpretability

☒ Transparency

☒ Correct

Correct! Transparency involves providing documentation for the system's decisions and actions (details about model architecture, training data, optimizations, etc.) to ensure compliance and accountability
2.

Why is explainability important for AI systems? (Check all that apply):

1 / 1 point

☒ Identifying Biases

☒ Correct

Correct! Explainable models can help uncover unintended biases, discrimination, or other ethical issues embedded in the training data or model.

☒ Trust and Accountability

☒ Correct

Correct! Being able to understand and audit an AI's decision-making process is crucial for building trust, especially in high-stakes domains like healthcare, criminal justice, and finance.

☒ Scientific Advancement

☒ Correct

Correct! Explainable models facilitate deeper understanding of the underlying phenomena being modeled.

☒ Human-AI Collaboration

☒ Correct

Correct! When humans can understand an AI's reasoning, they can provide guidance, feedback and corrections to improve future outputs. Debugging AI systems is part of this collaboration and explainable methods make debugging easier.
3.

What type of explanation is used to explain individual predictions of a black box machine learning model?

1 / 1 point

☒ Local Explanations

☐ Global Explanations

☐ Multimodal Explanations

☒ Correct

Correct! Local Explanations focus on explaining individual predictions of a black box machine learning model.

4.

In which explainability technique do you train a sparse linear model to mimic how the blackbox model predictions change with perturbations to the input instance?

1 / 1 point

☒ LIME

☐ Anchors

☐ Individual Condition Expectation

☒ Correct

Correct! Local Interpretable Model-agnostic Explanations (LIME) trains a sparse linear model to mimic how the blackbox model predictions change with perturbations to the input instance

5.

Which of the following is the CORRECT way to interpret Shapley Values?

1 / 1 point

☐ The Shapley value of a feature value is the difference of the predicted value after removing the feature from the model training.

☒ Given the current set of feature values, the contribution of a feature value to the difference between the actual prediction and the mean prediction is the estimated Shapley value.

☒ Correct

Correct! Given the current set of feature values, the estimated Shapley value represents the contribution of a particular feature to the difference between the actual prediction and the mean prediction.

6.

To better determine whether changes occur and in which direction(s) they occur for individual predictions in an ICE plot, using the following method can be beneficial:

1 / 1 point

☐ c-ICE

☐ PDP

☒ d-ICE

☒ Correct

Correct! The derivative ICE plot (d-ICE), which looks at individual derivatives of the prediction function with respect to the feature, can help us better determine whether changes occur and in what direction(s) they occur over a traditional ICE plot.

7.

How should I interpret a 2-way H-statistic being equal to 1?

1 / 1 point

☐ There is no interaction

☒ Each single partial dependence function is constant and the effect on the prediction only comes through the interaction

☐ All of the variance of the PDI_{jk} or f^* is explained by the sum of the partial dependence functions.

☒ Correct

Correct! The correct interpretation of the 2-way H-statistic being equal to one is that each single PD function is constant and the effect on the prediction only comes through the interaction.

8.

What challenge with Partial Dependence Plots (PDP) do Accumulated Local Effects (ALE) Plots attempt to solve?

1 / 1 point

☐ PDPs are complex and not very intuitive to implement.

☒ PDPs do not have clear interpretations when features are correlated

☐ PDPs only shows average marginal effects

☒ Correct

Correct! ALE plots are more robust when features are correlated than PDP plots.

9.

Fill in the blank: A

Counterfactual

○

 Explanation of a prediction describes the smallest change to the feature values that changes the prediction to a predefined output.

1 / 1 point

Response 1

☒ Counterfactual

☐ Prototype-based

☐ Influential Instances

☒ Correct

Correct! Counterfactual Explanations of a prediction describe the smallest change to the feature value that changes the prediction to a predefined output.

10.

What framework combines prototypes and criticisms?

1 / 1 point

☐ MMD-proto

☐ CPF (Criticism-Prototype Framework)

☒ MMD-critic

☒ Correct

Correct! MMD-critic, introduced in a 2016 paper, combines prototypes and criticisms in a single framework.